

Finite-Catalog Renewal Design: Small Menus and Resource-Deficit Paths

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Abstract. A renewal provider chooses a paid command book and allocates an accepted obligation across histories with expected caps and realization ceilings. We derive exact resource and price path representations with heterogeneous concave terminal rewards. Under a common reward, an optimum uses at most twice the number of distinct expected caps, and the bound is tight. Joint cap and reward classes admit the same reduction. Approximate reward classes yield an oscillation-based loss certificate without changing any feasibility constraint. Complementing edge resources merges all feasible anchors into one deficit dynamic program. Centered kernels and exact promise repair give an additive approximation with quadratic rather than cubic catalog dependence. Zero service costs yield exact lattice optimization at arbitrary command budgets; a saturated promise yields an exact all-budget frontier even with positive service costs. These regimes distinguish numerical granularity from binary encoding and complement weak SUBSET SUM hardness with a nonbinding budget. Price paths support finite branching and independent rational certificates. Synthetic experiments separate exact targets from positive tolerances, expose search mechanics, retain resource limits, and measure screening together with downstream solution and verification. A service-credit architecture distinguishes implementability from empirical calibration.

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1 Introduction

A renewal provider has accepted a total obligation but must implement it through a limited catalog of terminal commands. Different histories have different expected caps, realization ceilings, and costs of intermediate service. The provider chooses both the installed commands and the allocation of the obligation across histories. Opening charges discourage unnecessary commands. A useful algorithm must optimize these decisions jointly: changing the obligation or adding commands to repair an approximation changes the problem the provider agreed to solve.

This paper identifies two exact path representations of that joint problem. The first is a primal selected-boundary representation for a common increasing concave terminal reward. Every neighboring pair of selected commands owns a terminal-reward layer and the service costs of the histories whose eligibility ends at that boundary. These layers admit an explicit history-level reconstruction. Their capacities sum to the original available obligation, regardless of how the command book partitions histories. A uniform resource approximation can consequently be repaired at the original promise and command budget.

A history-level component-packing argument extends the primal representation to heterogeneous rewards by replacing a common chord with a concave within-edge allocation profile. The second representation is an exact priced path. It permits different increasing concave terminal rewards across histories. At a fixed obligation price, the optimal response of each history separates into its positive chord increments and its terminal service support. These terms can again be assigned to selected boundaries. One ordered dynamic program then optimizes all feasible anchors and command books. It computes a genuine original-space Lagrangian bound, not a bound on a gridded or relaxed-feasibility surrogate. The priced identity does not require different histories to share an ordering of marginal rewards; each history only needs its own concave ordering.

The complexity results distinguish menu structure from numerical encoding. The exact design decision problem is weakly NP-complete even with common unit linear reward, zero service costs, uniform history weights and a nonbinding command budget. A direct, encoding-sensitive subset-sum construction gives the proof; the obstruction is not inferred from the running time of a particular algorithm. In contrast, common expected caps admit an exact two-command reduction under a common terminal reward, even when the full catalog has many eligibility classes. Enumerating singletons and pairs solves this subclass in polynomial time without an accuracy parameter. It is distinct from the retained exact common-prefix subclass: neither identical caps nor identical eligibility is imposed by the general model. For general caps, rational price-path bounds support a finite, grid-free branch-and-bound algorithm. Every explored or unexplored leaf carries a valid upper bound, and every incumbent is an original-space lottery policy. The resulting interval is valid even when a run reaches its resource limit. The unrestricted exact search can still be exponential. The improvement is a polynomial node oracle and compact, data-dependent certificates, not an assertion of polynomial exact optimization in the NP-complete regime.

The common-cap result extends to a tight bound in the number of distinct expected caps. Within each cap class, a singleton or pair can preserve the class promise and improve its gross payoff. Taking their union yields an optimal book with no more than twice the number of cap classes. A matching construction requires every one of those commands. This gives an exact cap-count enumeration bound without identifying

fixed-exponent tractability with fixed-parameter tractability.

The capacity identity also leads to a different resource algorithm. Measure unused edge capacity instead of allocated obligation. Every feasible anchor then starts at zero deficit and every book ends at the same deficit. A single dynamic program replaces separately solved anchor problems, including for heterogeneous rewards. Centered concave kernels and exact repair give a polynomial additive guarantee with quadratic rather than cubic catalog dependence at prescribed accuracy. A zero-service lattice theorem further gives exact optimization for arbitrary command budgets after rational scaling. Under common quantity granularity this is a conventional pseudo-polynomial algorithm, explaining the encoding sensitivity of the subset-sum subclass. The manuscript separates the numerical size of the scaled lattice from its binary encoding length.

The path representation also strengthens preprocessing. A price path forced to contain a given command bounds every book using that command, including books in which it is the anchor. Comparing this bound with a feasible incumbent gives safe exclusion. This is complementary to the retained, inexpensive non-anchor envelope, which preserves targets uniformly but ignores the accepted promise and budget. Neither test guarantees a positive removal rate. We therefore measure exclusion rates, threshold slacks, missed exclusions, anchor removals, and total computation rather than identify an unchanged incumbent as proof that a new command is irrelevant.

The closest algorithmic antecedents are Lagrangian constrained-path methods, not a new form of generic branching. Handler and Zang (1980) combine dual bounds with path enumeration, and Beasley and Christofides (1989) use Lagrangian bounds for tree search and problem reduction. Fisher (1981) explains the general integer-programming role of such relaxations. Those mechanisms are used here, not claimed as inventions. The necessary additional theorem is that the uneliminated renewal problem has the stated exact priced arc functions, despite individual caps, pre-draw service, realization ceilings, and endogenous command choices. The proof also supplies a heterogeneous-reward extension that the common-reward primal rearrangement cannot provide.

The retained resource approximation relates to constrained-path discretization (Hassin, 1992; Lorenz and Raz, 2001), capped service allocation to separable convex resource allocation (Patriksson, 2008), and the uniform-grid acceleration to totally monotone matrix search (Aggarwal et al., 1987). Ordered quantizer design (Wu, 1991) and adjacent mean-preserving randomization (Pagès and Wilbertz, 2012) supply related ordering and interpolation ideas. They do not by themselves allocate a fixed aggregate obligation jointly with a paid, limited command book and history-specific realization restrictions. Our contributions concern the exact decompositions and their constructive connection to the original constraints, the tight cap-count compression, the merged-origin algorithm and exact lattice theorem, and the associated complexity distinctions. Gridding, dual weak bounds, and incumbent-based exclusion retain their established interpretations.

The computational study separates theorem validation from solver evaluation. Exact regressions compare against every feasible book on small models and reject altered certificates. A matched-limit study varies histories, catalog size, budget, accuracy, charges, and reward heterogeneity. It includes cases outside the allotted enumeration range, records interrupted runs, and reports proof-verification cost separately. Historical fine-accuracy cases are rerun rather than compared to timings from another machine. Additional zero-width runs expose anytime behavior, and near-threshold screening cases compare the two safe rules with exact

inclusion relevance. Every operational input remains synthetic. The paper makes a methodological case; it does not infer field calibration or a universal speed advantage from those experiments.

The new study executes the heterogeneous primal kernels, exposes exact-search mechanics, and compares screening plus downstream optimization and checking with direct solution. An operational service-credit architecture and an explicit payoff-calibration interface connect the model inputs to implementable decisions without describing synthetic instances as field data. Exact rational certificates distinguish correctness of the modeled decisions from the separate question of empirical model fit.

The constructive menu reduction also applies to finitely many joint cap and catalog-reward types. Conditioning the original promise on each joint class preserves heterogeneous rewards across classes and retains every history-level feasibility constraint.

Approximate reward types extend the menu guarantee beyond exact tariff equality. A catalog-wide reward-oscillation certificate bounds the loss from grouping rewards while leaving probabilities and every feasibility constraint unchanged. At the opposite end of the obligation range, a saturated promise fixes every individual target at its cap. The deficit representation then computes the exact menu-value frontier for all command budgets in one polynomial pass, even with heterogeneous rewards and positive service costs. These two results separate reward similarity, menu size and obligation slack as distinct sources of tractability.

2 Finite-Catalog Renewal Design

There are k histories. History j has probability $\pi_j > 0$, expected-obligation cap $b_j \in [0, 1]$, and realization ceiling $\tau_j \geq b_j$, with $\sum_j \pi_j = 1$. Write $\bar{b} = \sum_j \pi_j b_j$. Intermediate service has cost $h_j : [0, b_j] \rightarrow \mathbb{R}_+$, continuous, convex and nondecreasing, with $h_j(0) = 0$. The initial common terminal reward $f : [0, 1] \rightarrow \mathbb{R}$ is strictly increasing and concave. The catalog is $\mathcal{A} = \{a_1 < \dots < a_N\} \subset [0, 1]$. Opening command a_i costs $\rho_i \geq 0$.

A nonempty selected book $c = (c_1 < \dots < c_s)$ has $s \leq m$. After observing j , the provider selects deterministic pre-draw service $y_j \geq 0$ and a lottery C_j supported on c . The problem is

$$V^*(B) = \max_{c, y, C} \left\{ \sum_j \pi_j [\mathbb{E} f(C_j) - h_j(y_j)] - \sum_{z \in c} \rho(z) \right\}, \quad (1)$$

$$t_j = y_j + \mathbb{E} C_j \leq b_j, \quad y_j + C_j \leq \tau_j \quad \text{almost surely}, \quad \sum_j \pi_j t_j = B. \quad (2)$$

Here $B \in [0, \bar{b}]$ is accepted, rather than optimized away. The common-risk specialization is $\tau_j = b_j + \delta$. Expected and realized obligations are different restrictions. The terminal symbol does not carry a free history index, inherited service tier, or shared random seed. The cardinality m measures the command alphabet, not total implementation storage: targets, encoder logic, probability precision, program constants, and certificates are additional resources. The mathematical results below concern the stated finite catalog; continuous-design results have a separate approximation error.

For a feasible book, let $d_j = \max(c \cap [0, \tau_j])$, and let $F_{j,c}$ interpolate f linearly on this eligible prefix.

A singleton prefix has the constant value $f(d_j)$ at its only point.

Proposition 2.1 (Canonical response). *A book is feasible if and only if $c_1 \leq \min_j b_j$ and $c_1 \leq B \leq \bar{b}$. At target $c_1 \leq t \leq b_j$, its optimal response is*

$$v_{j,c}(t) = F_{j,c}(\min\{t, d_j\}) - h_j((t - d_j)_+). \quad (3)$$

It is attained by $y_j = (t - d_j)_+$ and an adjacent eligible lottery with mean $\min(t, d_j)$. The response is concave in t .

Proof. Every feasible command satisfies $C_j \leq \tau_j$ since $y_j \geq 0$. At mean $\mu = \mathbb{E}C_j$, concavity on the ordered support bounds the reward by $F_{j,c}(\mu)$. The payoff is at most $F_{j,c}(\mu) - h_j(t - \mu)$ for $c_1 \leq \mu \leq \min(t, d_j)$. Both terms are nondecreasing in μ , so the upper endpoint maximizes this bound. When $t \leq d_j$, adjacent interpolation has zero intermediate service and support below τ_j . When $t > d_j$, use command d_j surely and service $t - d_j$; the realized sum is $t \leq b_j \leq \tau_j$. Thus the bound is attained without weakening either restriction. Interpolation slopes decrease and are nonnegative; the tail slopes of $-h_j$ are nonpositive and nonincreasing. This proves concavity. Feasible targets fill the entire weighted interval $[c_1, \bar{b}]$, proving necessity and sufficiency. $\square$

The algorithmic specialization is

$$f(x) = rx - \frac{1}{2}qx^2, \quad r \geq q \geq 0, \quad r > 0, \quad h_j(y) = \frac{1}{2}\gamma_j y^2, \quad \gamma_j \geq 0. \quad (4)$$

All probabilities, caps, ceilings, catalog points, shape parameters, opening charges, and requested accuracies are rationally encoded in exact-arithmetic claims. Optimization identities allow real nonnegative charges; bit-complexity assertions do not assign a finite encoding to arbitrary real numbers. A common Lipschitz constant for the canonical responses is

$$L = \max\{r, \max_j \gamma_j\} > 0. \quad (5)$$

Continuity of the cost on its compact domain makes the stated extrema attained; every rational quadratic instance already satisfies this regularity. A fixed-book allocation can be solved by filling positive chord slopes in decreasing order, then zero-cost service capacity, then quadratic service tails. The service breakpoints are $\gamma_j(b_j - d_j)_+$. This takes $O(ks + k \log(k+1))$ rational operations with sorted marginal events. Alternatively, a direct supporting-price sweep verifies

$$\sum_j \pi_j v_{j,c}(t_j) = \lambda B + \sum_j \pi_j \max_{c_1 \leq t \leq b_j} \{v_{j,c}(t) - \lambda t\}. \quad (6)$$

Eligible knots, caps, and clipped stationary tail points suffice to check each scalar maximum. This certificate proves optimality for that book, not over all books.

3 Constructive Selected-Boundary Decomposition

This section assumes the common reward f in Section 2. For a selected book $c = (u_1 < \dots < u_s)$, write $a = u_1$ and $d_j = \max\{u_i : u_i \leq \tau_j\}$.

Lemma 3.1 (Global terminal-first allocation). *An optimal fixed-book canonical allocation cannot have both $t_i > d_i$ and $t_j < \min(b_j, d_j)$.*

Proof. Choose $0 < \delta \leq \min\{t_i - d_i, \pi_j[\min(b_j, d_j) - t_j]/\pi_i\}$. Reduce t_i by δ and increase t_j by $\pi_i\delta/\pi_j$. The weighted change is zero. The donor's terminal reward is unchanged and its service cost weakly decreases; the recipient's strictly increasing terminal interpolant improves. Both responses remain feasible by Proposition 2.1, contradicting optimality. $\square$

For selected neighbors $u < v$, define

$$\delta_j(u, v) = \mathbf{1}_{\{\tau_j \geq v\}} [\min(b_j, v) - u]_+, \quad T_{uv} = \sum_j \pi_j \delta_j(u, v), \quad (7)$$

$$J_{uv} = \{j : u \leq \tau_j < v\}, \quad Q_{uv} = \sum_{j \in J_{uv}} \pi_j (b_j - u)_+. \quad (8)$$

The capped service value on an exit set is

$$H_{uv}(y) = \min \left\{ \sum_{j \in J_{uv}} \pi_j h_j(z_j) : 0 \leq z_j \leq (b_j - u)_+, \sum_{j \in J_{uv}} \pi_j z_j = y \right\}, \quad 0 \leq y \leq Q_{uv}. \quad (9)$$

Empty sets have zero capacity and zero value at zero. Let $\sigma_{uv} = [f(v) - f(u)]/(v - u)$ and

$$\Phi_{uv}(x) = \sigma_{uv} \min(x, T_{uv}) - H_{uv}((x - T_{uv})_+), \quad 0 \leq x \leq T_{uv} + Q_{uv}. \quad (10)$$

For the terminal edge $(u, \dagger)$, use $T_{u\dagger} = 0$, $J_{u\dagger} = \{j : \tau_j \geq u\}$, and $\Phi_{u\dagger} = -H_{u\dagger}$. The functions Φ are concave. Under the quadratic primitives, they are L -Lipschitz with L as in (5).

Lemma 3.2 (Nested terminal layers). *Start each history at target a . Fill the selected terminal layers in increasing edge order. Once every preceding layer is full, each history with $\delta_j(u, v) > 0$ is at target u . Every weighted amount $x \in [0, T_{uv}]$ in the next layer has a feasible individual allocation. After all terminal layers are full, history j is at $\min(b_j, d_j)$; its remaining service capacity belongs to exactly one exit set.*

Proof. A history with positive increment on (u, v) has $b_j > u$ and $\tau_j \geq v$. It is eligible for all earlier selected commands, and the increments of those earlier layers telescope to $u - a$. This proves the first assertion, including a history whose cap will truncate the current layer.

For any desired x , initialize residual $r = x$ and visit histories in any fixed order. Give history j increment $e_j = \min\{\delta_j(u, v), r/\pi_j\}$ and reduce r by $\pi_j e_j$. Since $x \leq \sum_j \pi_j \delta_j(u, v)$, the final residual is zero. Each affected target lies in $[u, \min(b_j, v)]$ and is implemented by the adjacent lottery on u, v with upper probability $e_j/(v - u)$. It has no intermediate service and both realizations are below τ_j . Histories with zero increments keep their previously feasible responses.

If a layer is partial, stop: its predecessors are full and its successors unused. If every layer is full, the increments for history j telescope to $\min(b_j, d_j) - a$. It exits exactly after d_j , at an ordinary crossing edge or at the terminal edge. Thus its remaining capacity is $(b_j - d_j)_+$ and appears in exactly one H . Any feasible service vector on the exit sets implements target $\min(b_j, d_j) + z_j$. A positive z_j requires $b_j > d_j$, and its implementation is the sure command d_j plus service z_j ; their sum is at most $b_j \leq \tau_j$. Zero-capacity histories require no service. The exit sets are disjoint, so these implementations combine without conflicting allocations. $\square$

Lemma 3.3 (Implementable rearrangement). *For a fixed selected path, any feasible arc-resource vector can be rearranged, with unchanged total resource and no smaller payoff, into full earlier terminal layers, at most one partial terminal layer, and service only after all terminal layers are full. That rearranged payoff is attained by original history-level policies.*

Proof. Split the resource on each arc into its terminal and service parts in (10). If any terminal capacity remains unused while some service is positive, transfer the smaller of the available terminal capacity and a positive service amount to that terminal layer. The terminal gain is positive, while reducing service weakly lowers cost. Repeat until there is no such pair. This operation never changes charges or the sum of resources.

Next, if an earlier terminal layer is unfinished and a later one is positive, transfer the smaller of the unfinished capacity and the later mass. Selected chord slopes are positive and nonincreasing, so payoff cannot decrease. Filling layers successively yields the asserted form after finitely many exhausted donors or filled recipients. Each arc still respects its capacity: terminal mass is at most T_e , and service, when present, is at most Q_e after that arc's terminal part is full. Allocate any remaining service by minimizers of the corresponding H_e ; these minimizers exist by compactness and continuity. Lemma 3.2 implements the resulting terminal pattern and the disjoint service allocations. Its gross payoff is precisely $f(a) + \sum_e \Phi_e(x_e)$ for the rearranged vector. An arbitrary original vector need not be implemented at its unrearranged payoff; the weak improvement is the comparison needed for exact optimal values. $\square$

Theorem 3.4 (Exact selected-boundary resource path). *Let $P(c)$ contain the successive selected edges and the terminal edge of a feasible book. Then*

$$\sum_{e \in P(c)} (T_e + Q_e) = \bar{b} - a, \quad (11)$$

$$V_c(B) = f(a) - \sum_{z \in c} \rho(z) + \max_{\substack{0 \leq x_e \leq T_e + Q_e \\ \sum_e x_e = B - a}} \sum_{e \in P(c)} \Phi_e(x_e). \quad (12)$$

Maximizing over paths with at most m selected commands solves the original joint problem exactly. Reconstruction fills terminal layers in order, uses the greedy partial-layer allocation of Lemma 3.2, and then solves the disjoint capped service allocations. The resulting policy obeys the original promise, expected caps, realization ceilings, and command budget.

Proof. For history j , the terminal increments sum to $\min(b_j, d_j) - a$ and its unique service capacity is $(b_j - d_j)_+$. Their sum is $b_j - a$, proving (11) after weighting. In an original optimum, Lemma 3.1 places

service after terminal capacity. Among terminal layers, exchanging equal weighted amounts orders the common decreasing chord slopes; Lemma 3.2 justifies the individual feasibility of that ordering. If all terminal capacity is full, the exit-set cost minimizations describe all remaining feasible service allocations. Hence an original optimum supplies an arc allocation of the same value. Conversely, Lemma 3.3 implements a weakly improved version of every feasible arc allocation. The two comparisons yield equality of the attained maxima. Each selected charge is subtracted once, and the terminal edge does not open an additional command. $\square$

Corollary 3.5 (Endogenous eligibility). *A book of s commands has at most s nonempty selected-prefix classes. Inserting unselected candidates changes neither its arc functions nor its fixed-book optimum.*

Proof. The nonempty exit sets partition histories according to their last eligible selected command. Their definitions and terminal increments depend on the selected endpoints, not intervening unused candidates. $\square$

3.1 An eligibility-independent capacity potential

The path-capacity identity has a stronger local form. It explains why eligibility affects returns, but does not create a hidden resource state.

Proposition 3.6 (Capacity potential). *Define $p(u) = \sum_j \pi_j \min(b_j, u)$ for catalog vertices and $p(\dagger) = \bar{b}$. Every selected-boundary arc satisfies*

$$T_{uv} + Q_{uv} = p(v) - p(u), \quad Q_{u\dagger} = p(\dagger) - p(u). \quad (13)$$

Thus its capacity is independent of the realization ceilings as long as $\tau_j \geq b_j$. For a feasible anchor a , $p(a) = a$.

Proof. Fix one history and an ordinary edge $u < v$. If $\tau_j < u$, then $b_j < u$, so both its contribution to the arc and $\min(b_j, v) - \min(b_j, u)$ are zero. If $u \leq \tau_j < v$, then $b_j < v$ and its exit-service capacity is $(b_j - u)_+ = \min(b_j, v) - \min(b_j, u)$. If $\tau_j \geq v$, its terminal increment is $[\min(b_j, v) - u]_+$, which equals the same difference. These cases partition the histories. Multiplication by the probabilities and summation prove the first identity. For the terminal edge, $\tau_j < u$ again implies $(b_j - u)_+ = 0$, giving $Q_{u\dagger} = \sum_j \pi_j (b_j - u)_+ = \bar{b} - p(u)$. Finally, feasibility implies $a \leq b_j$ for every history. $\square$

Changing a ceiling can still change an arc's payoff through its exit set. Proposition 3.6 asserts capacity invariance, not value continuity across a threshold. It also identifies the renewal reduction as a capacity-conservative configuration of the type used in the production model below.

4 Menu Structure and Encoding Complexity

Equal expected caps do not require equal realization ceilings. A dense catalog may therefore give a different full-catalog prefix to every history even in the following exact regime.

Theorem 4.1 (Two commands suffice under common caps). *Assume $b_j = b$ for every history, the common increasing concave reward f , convex nondecreasing service costs, and nonnegative command charges. For any feasible $B \leq b$ and command budget $m \geq 1$, an optimal book exists with at most $\min(m, 2)$ commands. With rational quadratic primitives, enumerating feasible singletons and pairs and solving their fixed-book allocations uses*

$$O(N^2 k \log(k + 1))$$

rational arithmetic operations and polynomial bit complexity, independently of the number of eligibility classes and of any accuracy parameter.

Proof. The assertion is immediate for $m = 1$. Consider any feasible book with at least two commands and an optimal fixed-book allocation in the form of Lemma 3.2.

If the book contains B , the singleton $\{B\}$ is feasible. For any feasible policy, monotonicity and Jensen's inequality give

$$\sum_j \pi_j \{\mathbb{E}f(C_j) - h_j(y_j)\} \leq \sum_j \pi_j f(t_j) \leq f(B).$$

The singleton attains this upper bound at every target B and costs no more than a book containing it. It weakly improves that book.

If the largest selected command is $u < B$, all selected commands are below b and hence eligible for every history. All terminal layers can be filled up to u before service is used; the residual $B - u$ is service. The resulting targets are at least u . Deleting all other commands leaves the same terminal command, service allocation, targets, and gross value, while weakly reducing charges.

In the remaining case, let $u < B < v$ be the selected neighbors bracketing B . Every selected command through u is below b and eligible for every history. Completing the preceding layers puts every target at u . If $v \leq b$, the (u, v) layer has aggregate capacity $v - u$; the residual $B - u$ is smaller. The two-command book therefore implements the unique partial-layer pattern of the larger book, with no later terminal layer or service needed.

If $v > b$, histories admitting v have their cap below that command. They have no terminal increment beyond the (u, v) layer and no positive service capacity after it. Histories not admitting v admit no later selected command, so their last eligible selected command is u . For targets at least u , both groups have exactly the same canonical responses under the full book and under $\{u, v\}$. Later commands have zero useful capacity, and earlier commands have already been fully traversed. Deleting them preserves the optimized gross value. In all cases a singleton or pair weakly improves the original book after charges, proving the reduction.

There are at most $N + \binom{N}{2}$ retained possibilities. A two-command fixed-book response is computed by its history-level marginal events followed by capped convex quadratic service allocation, costing $O(k \log(k + 1))$ rational operations. The finite sums, sorted marginal ratios, and water-level equations have polynomial rational encoding lengths. Comparing the resulting rational values proves the arithmetic and bit claims. $\square$

The result is not a claim that every optimal book is small: additional zero-charge commands can remain in an equally good book. It also cannot be extended to arbitrary cap heterogeneity by the same argument.

With caps $1/4, 1/2, 3/4$, positive probabilities, $B = \bar{b}$, strictly concave common reward, zero service costs and charges, and a catalog containing these three caps, the gross Jensen upper bound $\sum_j \pi_j f(b_j)$ is attained by all three deterministic commands. Attaining it requires the respective commands; two cannot suffice. Conversely, even under a common cap, two commands can be necessary. With one history, $b = B = 1/4$, $\tau = 1/2$, catalog $\{0, 1/2\}$, zero costs and charges, and $f(x) = 2x - x^2/2$, the pair yields $7/16$ whereas the only feasible singleton yields zero.

The common-cap theorem and the retained common-prefix theorem in the electronic companion are different polynomial subclasses. Common caps allow many eligibility prefixes; common eligibility allows heterogeneous caps. Neither restriction is imposed on the price-path algorithm below. The next result shows why exact polynomial optimization cannot simply be extended to unrestricted heterogeneous caps unless $P = NP$.

4.1 NP-completeness with linear reward and free service

The exact common-cap theorem has a genuine boundary. Heterogeneous caps and paid command activation can encode a discrete target even when the reward and service primitives are linear and zero, respectively.

Theorem 4.2 (Complexity of exact finite-catalog design). *The rational quadratic joint-design decision problem, given a rational threshold ζ and asking whether $V^*(B) \geq \zeta$, is NP-complete. NP-hardness persists with the fixed common reward $f(x) = x$, zero intermediate-service costs, uniform probabilities, $\tau_j = b_j$, and a nonbinding command budget $m = N$.*

Proof. Membership in NP follows by supplying a selected book. Exact rational fixed-book allocation is computable in polynomial time and bit complexity, as in Proposition 5.2; its rational value can be compared with ζ . This argument applies to the full rational quadratic model, not to arbitrary unencoded convex functions.

For hardness, start with positive integer SUBSET SUM weights $w_1, \dots, w_n$, $n \geq 2$, total $A = \sum_i w_i$, and target $1 \leq W \leq A$. The NP-completeness of this binary-encoded problem is classical (Karp, 1972). Put

$$K = n^2 A, \quad \ell_i = \frac{i-1}{n}, \quad z_i = \ell_i + \frac{w_i}{nA}, \quad \pi_i = \frac{1}{n}, \quad b_i = \tau_i = z_i. \quad (14)$$

The catalog consists of the $2n$ distinct commands ℓ_i, z_i . Their order is $\ell_i < z_i < \ell_{i+1}$ for $i < n$, since positive weights and $n \geq 2$ imply $w_i < A$; also $z_n < 1$. Set $m = 2n$, charge zero for every ℓ_i , and charge $w_i/(2K)$ for z_i . Use $f(x) = x$ and $h_i = 0$. Define

$$D_0 = \frac{1}{n} \sum_i \ell_i, \quad B = D_0 + \frac{W}{K}, \quad \zeta = D_0 + \frac{W}{2K}. \quad (15)$$

These rational inputs have polynomial binary length. The root promise is feasible because $\bar{b} = D_0 + A/K$ and the minimum catalog command is zero.

Every feasible book can be augmented by all free commands ℓ_i without increasing its charge or exceeding the nonbinding budget. Enlarging its lottery support cannot reduce its attainable value. Hence an optimal

book contains all those commands. For an optional subset S of commands z_i , history i has last eligible command z_i when $i \in S$ and ℓ_i otherwise; higher free commands are above its realization ceiling. Its aggregate terminal capacity is

$$D(S) = D_0 + \frac{\sum_{i \in S} w_i}{K}.$$

Let $s = \sum_{i \in S} w_i$. With linear unit reward and free service, the optimized gross payoff is $\min\{B, D(S)\}$. For $B \leq D(S)$, lotteries between zero and each last eligible command provide every aggregate terminal mean up to $D(S)$. For $B > D(S)$, fill those terminal commands and use the remaining feasible service capacity $\bar{b} - D(S)$. Thus both parts of this gross-value formula are attained in the original constraints. Subtracting the optional charges gives the exact identity

$$V_S(B) = \min\{B, D_0 + s/K\} - \frac{s}{2K} = \zeta - \frac{|s - W|}{2K}. \quad (16)$$

Consequently $V^*(B) \geq \zeta$ holds if and only if a subset sums exactly to W . This is a polynomial reduction and proves NP-hardness under all the stated restrictions. $\square$

The reduction does not prove strong NP-hardness, hardness at a fixed small command budget, or a parameterized lower bound. It does establish that unrestricted exact polynomial-time optimization would imply $P = NP$. It is consistent with the additive scheme: a no-instance can be only $1/(2K)$ below the threshold, even though $f(x) = x$ has Lipschitz constant one. Resolving that distinction may require an inverse accuracy proportional to the numerical integer total. This identifies an encoding-sensitive barrier, not a proof that every instance requires the uniform grid or that its exact $1/\varepsilon$ dependence is optimal. The common-cap and common-prefix algorithms remain exact polynomial exceptions.

4.2 A tight cap-count bound for the command book

The common-cap result extends to a structural parameter without requiring common realization ceilings. Let $q = |\{b_j : 1 \leq j \leq k\}|$ be the number of distinct expected caps. The next result concerns a common terminal reward; heterogeneous rewards remain allowed in the resource and price algorithms below.

Theorem 4.3 (Cap-count compression). *Suppose the catalog is common and ordered, eligibility is induced by the realization ceilings, the terminal reward is common, strictly increasing and concave, service costs are continuous, convex and nondecreasing, and opening charges are nonnegative. Every feasible book and policy is dominated in value by a policy on a subset of that book containing at most $2q$ commands. Consequently some optimum uses at most $\min\{m, 2q\}$ commands. Given the original policy, the smaller book and a dominating policy can be constructed by solving at most q common-cap fixed-book allocation problems. For rational quadratic primitives this construction has polynomial arithmetic and bit complexity.*

Proof. Let I_h be a cap class, $W_h = \sum_{j \in I_h} \pi_j$, and let

$$B_h = \frac{1}{W_h} \sum_{j \in I_h} \pi_j t_j$$

be its conditional promise under the supplied policy. Every target is at least the original anchor, and B_h does not exceed the class cap. Treat the supplied book as the available catalog for class h , normalize its probabilities by W_h , and temporarily set opening charges to zero. Theorem 4.1 gives a singleton or pair c_h from that book whose optimized conditional gross payoff is no smaller than the supplied conditional payoff at B_h . This use of the theorem does not require the supplied class allocation to have been optimal.

For completeness, the book extraction is explicit. If B_h is selected, retain its singleton. If the largest selected command is below B_h , retain that command. Otherwise retain the two selected neighbors bracketing B_h . The common-cap proof verifies the original expected and realized constraints in each case, including a successor above the class cap.

Install $\tilde{c} = \bigcup_h c_h$ and retain each class's conditional policy. An enlarged available support does not invalidate a lottery that uses only its subgroup book. Every class keeps its exact conditional promise, so $\sum_h W_h B_h = B$ remains unchanged. Summing conditional gross payoffs proves gross dominance. Since $\tilde{c} \subseteq c$, nonnegative opening charges cannot increase. Also $|\tilde{c}| \leq \min\{|c|, 2q\}$. The fixed-book rational allocation and its encoding bound follow from Proposition 5.2 and the history-level marginal construction. An optimal supplied policy gives the existence claim. Existence itself follows from the finite book set and compact feasible allocations. $\square$

Proposition 4.4 (Tightness and exact enumeration). *The coefficient two in Theorem 4.3 is sharp for every positive integer q . For rational quadratic inputs, exhaustive evaluation through size $s = \min\{m, 2q\}$ is an exact algorithm with arithmetic count $O(N^s \text{ poly}(k, s))$ and polynomial bit cost per book. This is an XP bound in the cap count or command budget, not an FPT bound in either parameter alone.*

Proof. For tightness, take q histories of probability $1/q$. For $h = 1, \dots, q$, put

$$u_h = \frac{3h-2}{3q}, \quad v_h = \frac{3h-1}{3q}, \quad b_h = \frac{u_h + v_h}{2}, \quad \tau_h = v_h.$$

Use catalog $\{u_h, v_h : 1 \leq h \leq q\}$, zero service costs and charges, common reward $f(x) = 2x - x^2/2$, budget $2q$, and promise $B = \bar{b}$. The root equality and individual caps force every target to equal b_h . The full book attains the upper concave envelope of the eligible catalog at that target, with zero service. Strict increase rules out positive service at an optimum, since the eligible pair can attain terminal mean b_h . Strict concavity makes the two adjacent points u_h, v_h the unique supporting pair at this interior mean. Omitting either strictly decreases history h 's maximum payoff. All other histories are already at their individual maxima, so they cannot offset that decrease. Every optimum therefore uses all $2q$ commands.

For enumeration, evaluate every nonempty subset through size s whose anchor is feasible. Theorem 4.3 guarantees that this list contains an optimum, even if some original optimum contains extra zero-charge commands. There are $O(N^s)$ such books. The fixed-book marginal and capped-service calculation is polynomial in its rational input length. The unknown optimal allocation is not assumed available to this enumeration algorithm: the compression theorem justifies its search range. $\square$

These results distinguish three resources. A fixed small command budget gives exact polynomial enumeration with a budget-dependent exponent. A small cap count bounds the useful book, but does not

make that exponent independent of the cap count. Numerical lattice size gives a different exact algorithm, developed in Section 8.5, whose exponent does not grow with either parameter. Thus the nonbinding-budget hardness construction and the limited-menu results address different, explicitly stated input regimes.

4.3 Finite heterogeneous reward types

The compression argument is conditional, rather than intrinsically global. It therefore also covers finite heterogeneity in terminal rewards. Two histories have the same catalog reward type when their terminal reward values agree at every catalog command; behavior between commands does not affect a terminal lottery. Let g be the number of nonempty classes formed jointly by expected cap and catalog reward type. Service costs and prefix realization ceilings may vary within a class.

Theorem 4.5 (Joint cap–reward compression). *Suppose every terminal reward is strictly increasing and concave, service costs are continuous, convex and nondecreasing, and opening charges are nonnegative. Every feasible book and policy is weakly dominated by an original-feasible policy on a subset of that book containing at most $2g$ commands. In particular, an optimum exists with at most $\min\{m, 2g\}$ commands. Given the original policy, its compression uses at most g fixed-book allocation problems. For rational quadratic primitives, enumerating books of cardinality at most $\min\{m, 2g\}$ is exact with polynomial bit work per book. The exponent still depends on g or m ; this is an XP guarantee, not an FPT claim.*

Proof. Partition histories by their joint type. For class I , write $w_I = \sum_{j \in I} \pi_j$ and retain its conditional promise $B_I = w_I^{-1} \sum_{j \in I} \pi_j t_j$ from the given policy. The conditional policy is feasible under the class’s common cap, normalized positive probabilities, original ceilings and costs. At catalog commands its rewards equal one common strictly increasing concave reward. Its terminal-lottery objective consequently equals the common-reward objective, even if the rewards differ between commands. Apply Theorem 4.1 with the given book as the available catalog and zero charges for this conditional problem. It yields a singleton or adjacent pair with no smaller conditional gross payoff and the same B_I . The explicit bracketing construction in the proof of Theorem 4.3 supplies those commands before the fixed-book allocation is solved. Take the union of these conditional singleton/pair books. It is a subset of the original book, has at most $2g$ members, and does not exceed the original cardinality. Execute each class’s conditional policy without changing it. A union command assigned zero lottery probability in a class cannot violate that class’s ceiling. Conditional promises sum to $\sum_I w_I B_I = B$, so the original aggregate equality is unchanged. All history-level caps, pre-draw service requirements and realization ceilings remain satisfied. The gross payoff weakly increases and nonnegative opening charges cannot increase upon deletion. This proves dominance. There are at most g conditional allocations, followed, if desired, by reoptimization of the union book at B . Finite enumeration and the rational fixed-book allocation bound give the final assertion. $\square$

The common-reward cap-count theorem is recovered when the number of joint classes equals the number of distinct caps. Its strict $2g$ examples also show that the coefficient two cannot be improved uniformly over this larger model class. The bound does not promise compression when nearly every history has a different joint type. It identifies a useful finite-type regime without imposing common rewards on the heterogeneous resource algorithm.

4.4 Approximate reward types without changing feasibility

Exact reward classes are useful when service products share a tariff. Estimated reward curves need not coincide exactly. The following result gives a quantitative menu guarantee without rounding a cap, changing a probability, or relaxing a realization ceiling.

Partition histories into g groups, each with one common expected cap. For group h , choose a strictly increasing concave prototype $\widehat{f}_h$ on the catalog. The prototypes may differ across groups. Put

$$e_j(a) = f_j(a) - \widehat{f}_{h(j)}(a), \quad \Delta = \sum_j \pi_j \left[\max_{a \in \mathcal{A}} e_j(a) - \min_{a \in \mathcal{A}} e_j(a) \right]. \quad (17)$$

Only rewards are replaced in the prototype problem; probabilities, caps, ceilings, service costs, charges, promise and command budget remain original.

Theorem 4.6 (Oscillation-certified menu compression). *Every feasible original policy has a dominating prototype-policy compression on a subset of its book with at most $2g$ commands whose original value is at least the supplied original value minus Δ . In particular, some book of at most $\min\{m, 2g\}$ commands attains original value at least $V^*(B) - \Delta$. An optimal prototype book can be found by enumeration through that size; reoptimizing its allocation under the original rewards has the same guarantee. A prototype policy within ξ of the prototype optimum gives original regret at most $\Delta + \xi$.*

Proof. Write $J(P)$ and $\widehat{J}(P)$ for the original and prototype net values of the same feasible policy. Their difference is $E(P) = \sum_j \pi_j \mathbb{E}[e_j(C_j)]$; service costs and opening charges cancel. For any two feasible policies P, P' , each expectation lies between the catalog minimum and maximum of e_j , and hence $E(P) - E(P') \leq \Delta$. Apply Theorem 4.5 to the supplied policy in the prototype problem. Its construction keeps each group's conditional promise and returns a subset policy $\widetilde{P}$ with $\widehat{J}(\widetilde{P}) \geq \widehat{J}(P)$. It is feasible for the unchanged original constraints. Therefore $J(P) - J(\widetilde{P}) \leq E(P) - E(\widetilde{P}) \leq \Delta$. Apply this to an original optimum for existence. For the algorithmic statement, compare an original optimum P^* with a prototype ξ -optimal policy $\widehat{P}$; the inequality $\widehat{J}(P^*) - \widehat{J}(\widehat{P}) \leq \xi$ gives $J(P^*) - J(\widehat{P}) \leq \xi + \Delta$. Optimizing the original allocation on the recovered book can only improve it. Exact enumeration is justified by the prototype compression theorem, not by knowledge of P^* . $\square$

The bound is invariant to arbitrary history-specific additive reward constants. Thus exact equality of reward *increments* on the catalog suffices for zero-loss compression. If $|e_j(a)| \leq \epsilon_j$ for every command, then $\Delta \leq 2 \sum_j \pi_j \epsilon_j$; the oscillation can be strictly smaller. Choosing more prototype groups trades a larger guaranteed menu against a smaller certified reward loss. This does not assert tractable optimal clustering, nor does it merge distinct expected caps. With rational catalog values the certificate Δ takes $O(kN)$ rational operations; the enumeration exponent remains $\min\{m, 2g\}$. The additional regression compares original and prototype exact enumeration on identical feasibility inputs, including positive and zero reward dispersion.

5 Exact Price Paths with Heterogeneous Rewards

We now replace the common reward by a strictly increasing concave function f_j for each history. All other constraints and the common ordered catalog are unchanged. Proposition 2.1 holds history by history with interpolant $F_{j,c}$ of f_j . In particular,

$$v_{j,c}(t) = F_{j,c}(\min(t, d_j)) - h_j((t - d_j)_+)$$

is concave on the feasible target interval. The primal rearrangement in Section 3 need not hold across histories with differently ordered rewards. The following priced identity does not use that rearrangement.

For price $\lambda \in \mathbb{R}$, let

$$\sigma_{uv}^j = \frac{f_j(v) - f_j(u)}{v - u}, \quad K_j(u; \lambda) = \max_{0 \leq y \leq (b_j - u)_+} \{-h_j(y) - \lambda y\}, \quad (18)$$

$$\begin{aligned} A_{uv}(\lambda) &= \sum_{j: \tau_j \geq v} \pi_j [\min(b_j, v) - u]_+ (\sigma_{uv}^j - \lambda)_+ \\ &\quad + \sum_{j: u \leq \tau_j < v} \pi_j K_j(u; \lambda), \end{aligned} \quad (19)$$

$$A_{u^\dagger}(\lambda) = \sum_{j: \tau_j \geq u} \pi_j K_j(u; \lambda). \quad (20)$$

These supports are finite. At a feasible anchor a , define

$$Z_a(\lambda) = \sum_j \pi_j f_j(a) - \lambda a - \rho(a).$$

Theorem 5.1 (Exact priced selected-boundary identity). *For every fixed feasible book $c = (a = u_1 < \dots < u_s)$,*

$$\begin{aligned} &\sum_j \pi_j \max_{a \leq t \leq b_j} \{v_{j,c}(t) - \lambda t\} - \sum_{z \in c} \rho(z) \\ &= Z_a(\lambda) + \sum_{i=1}^{s-1} [A_{u_i u_{i+1}}(\lambda) - \rho(u_{i+1})] + A_{u_s^\dagger}(\lambda). \end{aligned} \quad (21)$$

Consequently, a longest ordered path with at most m selected vertices computes the exact maximum of the left side over all feasible books. Adding λB gives an upper bound on the original joint optimum, including under heterogeneous terminal rewards.

Proof. Fix a history and its eligible selected prefix. Beginning at a , its terminal interpolation consists of successive segments of lengths $[\min(b_j, v) - u]_+$, followed, when $b_j > d_j$, by its convex service-cost tail. The terminal chord slopes are positive and nonincreasing *within this history*. Subtracting λt therefore selects exactly all positive marginal terminal segments, with arbitrary selection on zero-slope ties. Their gain is the sum of segment length times $(\sigma_{uv}^j - \lambda)_+$.

For $\lambda \geq 0$, $h_j \geq 0$ and the feasible choice $y = 0$ show explicitly that $K_j(u; \lambda) = 0$. For $\lambda < 0$, every

positive terminal chord exceeds the price and is fully traversed before a service tail is selected. Maximizing the remaining concave tail gives $K_j(d_j; \lambda)$. Thus the complete scalar support is

$$f_j(a) - \lambda a + \sum_{\substack{(u,v) \text{ selected} \\ v \leq \tau_j}} [\min(b_j, v) - u]_+ (\sigma_{uv}^j - \lambda)_+ + K_j(d_j; \lambda).$$

The terminal terms are owned by their selected edges, and the last term by the unique exit edge or terminal edge of that history. Weighted summation gives (21), with each command charge subtracted once. The identity needs no comparison between chord slopes of different histories.

For an original feasible target vector, $\sum_j \pi_j t_j = B$ cancels the subtracted price when λB is added. Taking each scalar maximum and then maximizing the book proves weak duality. Every feasible book has a feasible anchor, so the ordered-path maximization omits none. $\square$

5.1 One recurrence for all anchors and restricted books

Let $\alpha = \min(B, \min_j b_j)$. At a fixed price, initialize

$$D_1(v) = Z_{a_v}(\lambda) \quad \text{when } a_v \leq \alpha, \quad D_1(v) = -\infty \quad \text{otherwise.}$$

For $\ell = 2, \dots, m$,

$$D_\ell(v) = \max_{u < v} \{D_{\ell-1}(u) + A_{a_u a_v}(\lambda) - \rho_v\}, \quad U(\lambda) = \lambda B + \max_{\ell, v} \{D_\ell(v) + A_{a_v \dagger}(\lambda)\}. \quad (22)$$

This recurrence handles every anchor simultaneously. The separate-origin resource baseline does not, whereas the deficit recurrence in Section 8 achieves the same source merge in primal coordinates. A predecessor pointer supplies a maximizing book. Its targets are recovered at the original promise by exact fixed-book allocation, not by adopting its generally infeasible priced target mean.

The same oracle permits a mandatory vertex set M and a forbidden set F . Remove F , disallow starting after the first mandatory vertex, crossing over a mandatory vertex, or terminating before the last mandatory vertex. If $|M| > m$ or $M \cap F \neq \emptyset$, the family is empty. Otherwise the allowed paths are exactly the books that contain M and avoid F : each mandatory vertex is either visited or would have to be skipped at the start, at an edge, or at the end. Prefix counts test these conditions in constant time per edge. Denote the resulting upper bound by $U_{M,F}(\lambda)$.

Proposition 5.2 (Rational quadratic price oracle). *Suppose $f_j(x) = r_j x - q_j x^2/2$ with rational $r_j \geq q_j \geq 0$, $r_j > 0$, and $h_j(y) = \gamma_j y^2/2$ with rational $\gamma_j \geq 0$. For a rational price, a restricted price path and its upper bound are computed in $O(kN^2 + mN^2)$ arithmetic operations after the input is read, with polynomial bit complexity. Exact fixed-book recovery costs $O(ks \log(ks + 1) + k \log(k + 1))$ operations for s selected commands.*

Proof. For $\gamma_j > 0$, the maximizing service is $\min\{(b_j - u)_+, \max(0, -\lambda/\gamma_j)\}$. When $\gamma_j = 0$, use zero service for nonnegative price and the full cap for negative price. Every support is a rational quadratic evaluation.

Computing all edge sums directly costs $O(kN^2)$; the recurrence uses mN^2 comparisons and additions. It is unaffected by the number of distinct full-catalog eligibility prefixes.

For recovery, sort the at most ks history-level terminal marginal segments by decreasing slope, preserving each history's internal order on ties. Fill them until the promise is met or all terminal capacity is used. Fill zero-cost service next and then solve the remaining capped quadratic water-level equation. This constructs an original feasible policy. Concavity of the individual responses supplies its supporting price, proving fixed-book optimality. All sorting keys and values are rational combinations of polynomially many input numbers. Exact common-denominator scaling for path comparisons and the resulting sums have polynomial bit lengths. The number of chosen search prices is not part of this single-oracle bound. $\square$

For a common quadratic reward, event-prefix sums accelerate repeated evaluations: $T_{uv} = p(v) - p(u) - Q_{uv}$ and the terminal term is $T_{uv}(\sigma_{uv} - \lambda)_+$. After sorting exit events for each predecessor, one price sweep accumulates service supports in $O(kN + N^2)$ operations before the $O(mN^2)$ path calculation. The reference implementation uses this optimization, exact integer-scaled comparisons, and a bounded cache of price kernels. Heterogeneous rewards use the direct edge sums. Arithmetic counts, rational bit complexity, and measured interpreter time are distinct quantities.

A small price gap does not imply strong duality over books. For one history, cap and ceiling one, $f(x) = x$, zero service cost, catalog $\{0, 1/2\}$, charges 0, $1/10$, and budget two, the optimized value is $\max\{0, \min(B, 1/2) - 1/10\}$. Its slope increases at $B = 1/10$. At $B = 1/20$ the true optimum is zero but its concave dual majorant is positive. The next section closes such gaps by a complete discrete partition rather than silently treating a single price as a global optimality certificate.

5.2 A heterogeneous-reward primal extension

The priced identity also suggests the correct generalization of the primal arc payoff. Simply replacing the common reward by different rewards in (10) would be incorrect, because one chord slope no longer describes all histories. Instead, define the within-edge terminal allocation value

$$R_{uv}(x) = \max \left\{ \sum_{j: \tau_j \geq v} \pi_j \sigma_{uv}^j e_j : 0 \leq e_j \leq \delta_j(u, v), \sum_j \pi_j e_j = x \right\}, \quad 0 \leq x \leq T_{uv}. \quad (23)$$

The feasible within-edge polytope is nonempty and compact at each stated total, so its linear objective attains a maximum. Its value is continuous, increasing and concave, with marginal slopes drawn from the history-specific chord slopes. Set

$$\tilde{\Phi}_{uv}(x) = R_{uv}(\min(x, T_{uv})) - H_{uv}((x - T_{uv})_+). \quad (24)$$

The terminal edge still has payoff $-H_{u\ddagger}$. All terminal slopes are positive and all service marginal returns are nonpositive, so this joined payoff is concave.

Theorem 5.3 (History-level component packing). *For a common ordered catalog with ceiling-induced prefix eligibility and heterogeneous increasing concave terminal rewards, replace $f(a)$ in (12) by $\sum_j \pi_j f_j(a)$ and*

every Φ by $\tilde{\Phi}$. The resulting scalar-resource path identity is exact. Its capacities remain those in (11). For any feasible arc-resource vector, within-edge optimizers followed by history-level packing produce a feasible original policy with the same individual component totals and no smaller payoff.

Proof. For a fixed book, introduce a component $e_{j,uv} \in [0, \delta_j(u, v)]$ for every eligible terminal segment, and service $z_j \in [0, (b_j - d_j)_+]$. Temporarily drop the requirement that a history's earlier components must be filled before its later ones. The relaxed objective, apart from charges, is

$$\sum_j \pi_j \left[f_j(a) + \sum_{(u,v)} \sigma_{uv}^j e_{j,uv} - h_j(z_j) \right], \quad \sum_j \pi_j \left[\sum_{(u,v)} e_{j,uv} + z_j \right] = B - a.$$

Every original canonical policy embeds in this relaxation by filling its eligible segments in order and then its service tail. Conversely, fix any relaxed component allocation and define $t_j = a + \sum_{(u,v)} e_{j,uv} + z_j$. The sum of that history's component capacities is exactly $b_j - a$, so $a \leq t_j \leq b_j$. Its components can be packed in their original order while preserving t_j : move service into any unfinished positive terminal segment, then move mass from a later positive segment into any unfinished earlier segment. Each move weakly improves payoff, because the history's terminal slopes are decreasing and positive and service cost is nondecreasing. This comparison is entirely within a history; it needs no common ordering across histories. At termination the components implement its canonical response at t_j . Proposition 2.1 supplies the adjacent lottery or sure-command service policy, with every realization constraint intact. The weighted promise is unchanged by packing. Thus relaxation and original problem have the same optimum.

Now regroup the relaxed components by selected edge, assigning z_j to the unique exit edge of history j . Groups share only the scalar total-resource equality; their individual component bounds are otherwise disjoint. For a fixed group resource x , optimal allocation fills positive terminal returns first, whose optimum is R_{uv} , and then minimizes the exit-set service cost H_{uv} . This is exactly (24). Maximizing over group resources is therefore the same relaxation already proved exact. The component-capacity telescoping is unchanged. Within-edge optimizers followed by the preceding history-level packing provide the asserted constructive recovery. $\square$

In the quadratic specialization, R_{uv} is evaluated by sorting the eligible history slopes and filling weighted segment capacities. Its Lipschitz constant is at most $\max_j r_j$. The service part is bounded by $\max_j \gamma_j$. Hence the polynomial additive resource construction also extends to heterogeneous rewards with these new kernels. The retained uniform-grid implementation used in the comparison study has only the common-reward kernels; it is not represented as an executed heterogeneous solver. The new price-path implementation and independent certificates do support heterogeneous rewards. Those statements describe the antecedent comparison. The merged-origin implementation in Section 8 now evaluates these heterogeneous primal kernels directly; the new study tests and independently checks them rather than carrying the extension only as a theorem.

6 Finite Exact Search and Original-Instance Certificates

6.1 Price bounds, feasible policies, and discrete coverage

For a restricted family (M, F) , every evaluated price yields an upper bound $U_{M,F}(\lambda)$. Recovering the price-maximizing book at B gives a feasible lower value. Neither the priced mean nor a solver's completion status is a feasible-policy witness. Maintain a global incumbent $\underline{V}$ and, for each live leaf, its best completed price bound. The gross terminal-reward bound $\sum_j \pi_j f_j(1)$, dropping nonnegative service costs and opening charges, initializes a leaf before its first price evaluation.

The reference search evaluates a bounded number of prices at each node. The target range of a maximizing book gives a valid subgradient direction for the convex price bound. A bracket starts at $-\max_j \gamma_j$ and $\max_j r_j$; candidate prices also include the exact supporting price of the recovered fixed-book policy. Bisection supplies a distinct rational price when a candidate repeats. Bracketing is a search heuristic, not a premise for validity: every completed price produces a valid upper bound independently of how it was chosen.

If a node bound is at most $\underline{V} + \varepsilon$, it need not be split. Otherwise select an undecided candidate and create the two children in which it is respectively required and forbidden. Both inherit the parent's bound until a sharper completed calculation is available. Mandatory and forbidden vertices are enforced by the recurrence in Section 5. A family with one possible book is evaluated at that book's exact supporting price. Every feasible original book remains in exactly one leaf family.

Theorem 6.1 (Grid-free finite exact algorithm). *Under the rational quadratic assumptions of Proposition 5.2, the preceding algorithm returns, after every completed node or price calculation, an original feasible policy and a rational interval*

$$\underline{V} \leq V^*(B) \leq \bar{V}, \quad \bar{V} = \max\{\underline{V}, U_{M,F} : (M, F) \text{ is a current leaf}\}. \quad (25)$$

An empty leaf is omitted. With no resource limit and zero requested width, finite binary branching and exact singleton-family evaluation terminate at $\underline{V} = \bar{V}$. The full binary cover has at most $2^{N+1} - 1$ nodes. A node may use multiple prices, so total nodes and completed price calls are different counts. Each completed price oracle has the polynomial complexity of Proposition 5.2. At positive tolerance, or under interruption, the actual width $\bar{V} - \underline{V}$ is a valid instance-dependent additive error bound; it has no mandatory inverse-tolerance resource grid.

Proof. The initial incumbent is an original feasible singleton at the minimum command. Every subsequent incumbent is recovered at B and respects the same constraints. Theorem 5.1 bounds every book in a leaf. Requiring or forbidding a previously undecided candidate partitions its family exactly, so inherited bounds remain valid on both children. A node with no allowed path is empty; all nonempty leaf families have feasible target allocations by Proposition 2.1. These facts prove (25), including leaves not yet processed or calculations interrupted before completion.

Along a root-to-leaf chain, a candidate is decided at most once. After at most N splits the family is empty or fixes one book. Concave fixed-book allocation has a supporting price at B (at a boundary promise, a one-sided supergradient extended to the feasible interval suffices, and the quadratic model has

finite marginal bounds), and its scalar maximizers attain the price bound at that same promise. Consequently the restricted priced optimum at that price equals the recovered value. A fixed finite number of price attempts per nonterminal node followed by an undecided split cannot prevent this finite termination. The elementary depth- N binary-tree count supplies the worst-case bound. Rational formulas, finite supports, and original-space policies establish exact arithmetic throughout. $\square$

The worst-case exponential bound is deliberate. Theorem 4.2 proves NP-completeness of the unrestricted decision problem. The present algorithm supplies a finite exact method with a polynomial node relaxation and a data-dependent certified stopping rule; it does not assert a polynomial worst-case exact running time. The exact common-cap and common-prefix theorems give polynomial subcases. For either reward model with the appropriate arc kernels, the uniform-resource scheme in Section 7 retains a separate worst-case polynomial additive guarantee. A valid combined interval can take the maximum of the feasible lower values and the minimum of the price-tree and resource-grid upper bounds. No assumption of strong duality across books is required.

6.2 Independent verification and proof economics

A price certificate contains the original model, its catalog and charges, a feasible lottery policy, and the complete binary coverage tree. A priced leaf records its rational price and upper bound; a still-unprocessed leaf may retain the universal bound. The independent checker reconstructs arc supports by direct endpoint and stationary-point maximization, then uses a backward at-most-budget recurrence rather than the optimizer's forward exact-length recurrence. It checks every branch, every mandatory or forbidden restriction, and every leaf bound. A parent price bound may be looser than a child's recomputed bound; the required comparison is dominance, not equality.

The lower witness retains nonnegative service, lottery masses, command support, every positive-probability realization ceiling, all expected caps, and the exact weighted promise. The checker subtracts the original selected charges and recomputes the global interval. A separately supplied instance digest can bind the proof to the requested input; a self-reported digest alone is not authentication. The new checker raises explicit errors and does not rely on assertions being enabled. The inherited grid checker continues to require ordinary Python execution, as documented in its retained source.

The proof record stores $O(k + s + N + P)$ rational fields and tree entries up to the history-level lottery representation, where P is the number of retained nodes. It need not serialize the scalar-resource tables. Verification recomputes polynomial price paths at the leaf prices; it is not free, and a large branching tree can still be expensive. Input and price bit lengths contribute to actual byte counts. The study reports compressed and uncompressed size, checking time, peak process memory, and resource-limit outcomes separately from the command alphabet.

6.3 Incumbent-based exclusion, including anchors

The uniform anchor-envelope rule retained in the electronic companion cannot delete a possible anchor. The priced oracle provides a complementary test.

Proposition 6.2 (Forced-command price exclusion). *Let v_0 be the value of an original feasible incumbent. For any candidate z , including a possible anchor, let $U_z(\lambda)$ be the price-path upper bound with z mandatory. If $U_z(\lambda) < v_0$, no optimal book contains z . All commands certified by such strict inequalities may be removed simultaneously without changing the optimal value. Under a weak inequality, simultaneous deletion is safe only with an independently retained incumbent avoiding all deleted commands.*

Proof. Every book containing z lies in its forced family and has value at most $U_z(\lambda)$. A strict comparison excludes it from optimality and also implies that the incumbent does not contain it. Therefore every optimal book and the incumbent avoid the union of strictly excluded candidates. The reduced problem retains an optimal policy and introduces no new one. For weak inequalities, any book lost by a deletion has value at most v_0 , while an incumbent avoiding the entire deletion set attains v_0 in the retained problem. This proves that qualified version as well. $\square$

This is the familiar incumbent-based Lagrangian reduction principle (Beasley and Christofides, 1989; Fisher, 1981), applied through an exact original-instance price oracle. Its benefit here is that the bound incorporates the accepted promise, command budget, and other selected alternatives. No uniform removal fraction follows. Reusing the arc supports for one price, evaluating every forced candidate costs $O(mN^3)$ additional arithmetic operations. The cheap $O(kN)$ non-anchor envelope remains a different option with an order-independent target-box guarantee. The screening study reports both costs and compares their exclusions with exact inclusion relevance rather than declaring one rule universally preferable.

7 A Polynomial Additive Fallback

For the common-reward model, or the heterogeneous kernels of Theorem 5.3, the selected-boundary representation yields a complementary worst-case guarantee. An arc has concave L -Lipschitz payoff and capacity $T_e + Q_e$. Every book with feasible anchor a has capacity $\bar{b} - a \geq B - a$. Round each of its at most m arc resources down to multiples of η , including the terminal edge. Accept total resource in $[B - a - m\eta, B - a]$. Every accepted path has enough spare capacity to restore its missing resource without changing its selected commands.

Theorem 7.1 (Original-promise additive fallback). *For rational increasing concave quadratic rewards and quadratic service costs, let $L = \max_j \{r_j, \gamma_j\}$ and set $\eta = \varepsilon / (2Lm)$ and let S_η be the best accepted grid-path value over all feasible anchors. Exact fixed-book recovery at B yields*

$$\underline{V} \leq V^*(B) \leq S_\eta + Lm\eta, \quad S_\eta + Lm\eta - \underline{V} \leq \varepsilon.$$

With $Q = \lceil 2Lm/\varepsilon \rceil + 1$, direct kernel construction and resource convolution have arithmetic bound

$$O(kN^2Q \log(k+1) + mN^3Q \log(Q+1)), \quad (26)$$

and bit complexity polynomial in the numerical grid size and binary input length.

Proof. Rounding the resources of an optimal path loses at most $Lm\eta$ in payoff. Conversely, repairing a maximizing accepted path changes its coordinates upward by total at most $m\eta$ and loses at most $Lm\eta$. The constructive identity implements the repaired vector with no smaller payoff; exact fixed-book recovery can improve it further. This proves the interval. Precompute each of $O(N^2Q)$ arc values by sorted terminal components and capped quadratic service allocation. At a fixed edge, a concave grid kernel gives an anti-Monge max-plus convolution even if the previous Bellman values are not concave. Monotone row search costs $O(Q \log(Q+1))$ per transition. Counting anchors and command lengths gives (26). The full recurrence, rational scaling argument, and independent checker are retained in the electronic companion. $\square$

Writing normalized accuracy as $\delta = \varepsilon/L$ exposes the effective terms

$$O\left(\frac{kmN^2}{\delta} \log(k+1) + \frac{m^2N^3}{\delta} \log(m/\delta + 2)\right).$$

Thus polynomial arithmetic does not by itself imply practical scalability. At a prescribed absolute error the dependence is on L/ε , not its binary encoding length. Signed net payoffs and values close to zero preclude a general multiplicative interpretation. The price-tree method instead stops at its observed original-instance width, sometimes at exact equality, but has an exponential worst-case discrete tree. These are complementary guarantees, not a claim that the uniform $1/\varepsilon$ dependence is necessary. The broader heterogeneous-reward guarantee uses the newly derived within-edge profiles, not an unchanged common-reward kernel. The inherited software used as the uniform baseline remains restricted to common rewards.

8 Merged-Origin Resource-Deficit Paths

The capacity potential does more than guarantee repair after rounding. It identifies a resource coordinate in which every feasible anchor has the same initial state and every feasible book has the same terminal state. This removes the extra anchor loop in the resource algorithm. The construction applies to the heterogeneous kernels of Theorem 5.3.

8.1 An exact source-independent coordinate

Write

$$p(u) = \sum_j \pi_j \min\{b_j, u\}, \quad p(\dagger) = \bar{b}, \quad C_{uv} = T_{uv} + Q_{uv} = p(v) - p(u).$$

For the terminal edge, $C_{u\dagger} = \bar{b} - p(u)$. Every feasible anchor $a \leq \min\{B, \min_j b_j\}$ satisfies $p(a) = a$. If edge e is assigned resource x_e , define its unused capacity $w_e = C_e - x_e$ and its deficit payoff $\Psi_e(w) = \Phi_e(C_e - w)$. Opening charges are kept separate from Φ_e throughout. Let $R = \bar{b} - B$.

Theorem 8.1 (Merged-origin identity). *Under the common ordered catalog, prefix eligibility, and increasing*

concave history-specific rewards of Theorem 5.3, the original value equals

$$\max_{\substack{c: 1 \leq |c| \leq m \\ a = \min c \leq \min(B, \min_j b_j)}} \left\{ \sum_j \pi_j f_j(a) - \sum_{z \in c} \rho(z) + \max_{\substack{0 \leq w_e \leq C_e \\ \sum_{e \in P(c)} w_e = R}} \sum_{e \in P(c)} \Psi_e(w_e) \right\}. \quad (27)$$

All anchors start at cumulative deficit zero and end at cumulative deficit R . Every feasible deficit vector on a chosen book yields a policy at the original promise, within the original command budget, with at least its displayed value. No anchor identifier is needed in a Bellman state once the current command and number of selected commands are known.

Proof. For a fixed book with anchor a , capacity conservation gives $\sum_e C_e = \bar{b} - a$. Hence

$$\sum_e x_e = B - a \iff \sum_e (C_e - x_e) = \bar{b} - B = R.$$

The transformation is a bijection between the two resource boxes intersected with their equalities. Apply the exact component-packing identity to obtain (27) and its recovery claim. Along a prefix ending at u , the cumulative deficit is

$$d = p(u) - a - \sum_{e \text{ in the prefix}} x_e.$$

At the anchor it is zero; traversing $u \rightarrow v$ adds $C_{uv} - x_{uv}$. Continuation payoffs and transitions depend only on u , the remaining command allowance, and d , not on the earlier anchor. A Bellman merge retains the best accumulated value among such prefixes. The recovered full path retains its own anchor through predecessor pointers. Partial paths need not themselves encode a canonical history policy: the full-path packing theorem supplies implementation after the equality is restored. $\square$

The same argument gives an abstract reuse principle. On an acyclic graph, suppose every resource capacity has the form $C_{uv} = p(v) - p(u) \geq 0$, all paths share a sink, and a source a requires resource $B - p(a)$ with $p(a) \leq B$. Complementing edge resources makes the total unused capacity $p(\dagger) - B$ independent of the source. Any additive path objective and edge-count bound then admit the same source merge. Concavity is needed for the approximation and fast convolution below, not for this algebraic identity. The renewal-specific content is the exact derivation of those capacities and payoffs from individual expected and realized constraints; an arbitrary constrained path need not have this potential structure.

8.2 Centered approximation and exact repair

For rational quadratic primitives put

$$r_{\max} = \max_j r_j, \quad G = \max_j \gamma_j, \quad \lambda_0 = (r_{\max} - G)/2, \quad K = (r_{\max} + G)/2 > 0.$$

Every supergradient of Φ_e lies between $-G$ and $r_{\max}$ on its interval. Consequently $g_e(w) = \Psi_e(w) + \lambda_0 w$ is concave and K -Lipschitz. Adding this common linear term changes every exactly feasible path value by the

same constant $\lambda_0 R$; it does not change the optimizer.

For $\varepsilon > 0$ and $R > 0$, set

$$\eta = \frac{\varepsilon}{2Km}, \quad Q = \left\lceil \frac{R}{\eta} \right\rceil + 1, \quad \mathcal{I} = \{s \in \{0, \dots, Q-1\} : R - m\eta \leq s\eta \leq R\}. \quad (28)$$

Let $D_{\ell,v}(s)$ denote the best accumulated centered value at command a_v , with exactly ℓ selected commands and cumulative deficit $s\eta$. Unreachable entries are $-\infty$. All feasible anchors are initialized simultaneously:

$$D_{1,i}(0) = \sum_j \pi_j f_j(a_i) - \rho_i, \quad a_i \leq \min\{B, \min_j b_j\}, \quad (29)$$

$$D_{\ell,v}(s) = \max_{\substack{u < v, 0 \leq t \leq s \\ t\eta \leq C_{uv}}} \{D_{\ell-1,u}(s-t) + g_{uv}(t\eta) - \rho_v\}, \quad (30)$$

$$S_\eta = \max_{\substack{1 \leq \ell \leq m, u, s \in \mathcal{I} \\ 0 \leq t \leq s, t\eta \leq C_{u^\dagger}}} \{D_{\ell,u}(s-t) + g_{u^\dagger}(t\eta)\}. \quad (31)$$

The tail does not install a new command. A book with s selected commands has s resource edges including that tail.

Theorem 8.2 (Centered deficit approximation). *Equations (29)–(31) and exact fixed-book recovery produce an original-instance interval*

$$\underline{V} \leq V^*(B) \leq S_\eta - \lambda_0 R + Km\eta, \quad S_\eta - \lambda_0 R + Km\eta - \underline{V} \leq \varepsilon. \quad (32)$$

The guarantee allows heterogeneous rewards, arbitrary full-catalog eligibility classes, arbitrary nonnegative charges, and the original budget. For $R = 0$ the same recurrence uses only deficit zero and is exact, including with nonzero service costs.

Proof. Round each deficit on an optimal continuous path down to a multiple of η . Its cumulative rounded deficit is no greater than R and differs from R by less than $m\eta$. Every prefix stays in the grid. Lipschitz continuity loses at most $Km\eta$ from the centered optimal value $V^*(B) + \lambda_0 R$. Thus $S_\eta \geq V^*(B) + \lambda_0 R - Km\eta$, proving the upper bound.

Conversely, recover a maximizing grid path. Increase its deficits, without exceeding their individual capacities, until their sum is R . This is always possible because the path's total capacity is $\bar{b} - a \geq \bar{b} - B = R$. The total increase is at most $m\eta$. The centered payoff decreases by at most $Km\eta$; after subtracting $\lambda_0 R$, the uncentered path is worth at least $S_\eta - \lambda_0 R - Km\eta$. Component packing gives an original feasible policy with no smaller value. Exact fixed-book allocation may improve it further, so the resulting feasible lower value satisfies the same bound. This proves (32). When $R = 0$, nonnegative deficits must all be zero, so there is no rounding, repair, or approximation error. $\square$

8.3 Convolution with holes and bit complexity

The merge can create unreachable gaps in a Bellman row. Concavity of the row itself is neither assumed nor generally true.

Lemma 8.3 (Concave convolution with arbitrary finite support). *Let $A(z) \in \mathbb{R} \cup \{-\infty\}$ be arbitrary on $0, \dots, Q-1$, and let $g(t)$ be a finite concave sequence on $0, \dots, L$. The smallest maximizing input index in*

$$M(s) = \max_{\max(0, s-L) \leq z \leq s} \{A(z) + g(s-z)\}$$

is nondecreasing across rows having at least one finite feasible column. All values and predecessors can be computed in $O(Q \log(Q+1))$ comparisons and additions, including arbitrary holes in the support of A .

Proof. Delete columns with $A(z) = -\infty$ and rows whose feasible interval contains no remaining column. Concavity gives the anti-Monge inequality

$$[A(z) + g(s-z)] + [A(z') + g(s'-z')] \geq [A(z') + g(s-z')] + [A(z) + g(s'-z)]$$

whenever $s < s'$, $z < z'$, and the four entries are feasible. If two maximizing indices crossed in the wrong order, their crossed rectangle would be feasible because $0 \leq s-z \leq L$ is a staircase band. The inequality and the smallest-index tie rule would contradict that crossing. The maximizing indices are therefore monotone. Sorted finite columns permit feasible-row detection by interval searches. Divide-and-conquer row maximization scans a total $O(Q)$ candidate positions at each of $O(\log(Q+1))$ levels, with shared endpoint candidates included. Empty rows are discarded before recursion; they are not allowed to impose an artificial predecessor bound on another row. $\square$

Proposition 8.4 (Arithmetic, storage, and verification). *For rational quadratic inputs the centered deficit algorithm has arithmetic bound*

$$O(kN^2Q \log(k+1) + mN^2Q \log(Q+1)). \quad (33)$$

Its cached implementation stores $O((N^2 + mN)Q + kN^2)$ rational entries and has bit complexity polynomial in binary input length and numerical Q . The extra factor N for separately enumerated anchors in (26) is absent. An independent checker can reconstruct all kernels and verify Bellman upper-potential inequalities in $O(k^2N^2Q + mN^2Q^2)$ rational operations, without using the optimizer's monotone search.

Proof. There are $O(N^2)$ ordinary and terminal edges. Sorted weighted terminal segments and sorted service saturation thresholds give piecewise-linear terminal profiles and piecewise-quadratic service profiles. Their endpoints and cumulative values are rational. Direct evaluation by capped allocation already gives the first term in (33); caching endpoints and binary searching them can reduce its constant and history dependence. For each selected count and edge, Lemma 8.3 gives the second term. The predecessor table has mNQ entries. Caching every edge grid adds N^2Q entries; retaining all component descriptions adds at most kN^2 . Thus the storage claim includes the edge cache rather than counting Bellman states alone.

A kernel value involves finite sums and products of input rationals, a grid rational, and at most one capped-service water-level division. Sums of at most k rational reciprocals have encoding length polynomial in the total input encoding. A path combines at most m such values. Therefore all represented integers have bit length polynomial in the input length and $\log(Q+1)$; multiplying arithmetic counts by polynomial

integer-arithmetic costs gives the stated bit bound. A common denominator may have exponentially large numerical value, which is not a polynomial-time claim in its binary encoding.

For independent verification, supplied Bellman values must dominate every feasible anchor and every predecessor-plus-edge transition. No transition into a claimed unreachable entry may be omitted. Every accepted terminal combination must be covered by the reported terminal bound. These inequalities inductively bound every grid path; an attaining path certifies the grid optimum used in the repair calculation. A separately checked original lottery supplies the lower value. Direct service-threshold testing costs at most quadratic time in histories per kernel value; explicitly examining every predecessor deficit gives the stated conservative checker bound. The checker uses neither the forward optimizer nor its divide-and-conquer routine. $\square$

With $\delta = \varepsilon/K$ and $R \leq 1$, substitution gives a budget-quadratic, catalog-quadratic term of order $m^2 N^2 \delta^{-1} \log(m/\delta + 2)$ rather than its catalog-cubic predecessor. The guarantee is additive in signed net value. It is not polynomial in $\log(1/\varepsilon)$, and a successful optimization need not imply a cheap independent verification; both costs are measured separately.

8.4 An exact all-budget frontier at a saturated promise

The zero-deficit case is a second exact regime, distinct from zero service cost. It permits fully heterogeneous caps, rewards and strictly positive service costs.

Corollary 8.5 (Saturated-promise menu frontier). *For rational quadratic primitives and $B = \bar{b}$, all optimum values for command budgets $1, \dots, N$ and an attaining book for each budget can be computed together in $O(kN^2 + N^3)$ rational operations with polynomial bit complexity. Storing edge weights, Bellman values and predecessors requires $O(N^2)$ rational or integer entries. No common lattice or reward-type assumption is needed. The original policy is recovered at the unchanged targets $t_j = b_j$.*

Proof. Since probabilities are positive and $t_j \leq b_j$, the equality of weighted sums forces $t_j = b_j$ for every history. In the deficit representation $R = 0$, all nonnegative edge deficits are zero. The full-capacity edge weight is

$$w_{uv} = \sum_{j: \tau_j \geq v} \pi_j \sigma_{uv}^j \delta_j(u, v) - \sum_{j \in J_{uv}} \pi_j h_j((b_j - u)_+) - \rho(v),$$

with the first sum and opening charge omitted on the sink edge. Computing each weight directly costs $O(k)$ operations: all terminal components and all service components are full, so no sorting or water-level search is required. Initialize every feasible anchor a_i with value $\sum_j \pi_j f_j(a_i) - \rho_i$. For each exact selected count, propagate these weights along the ordered graph. All N layers cost $O(N^3)$ operations. Adding the sink weights and taking cumulative maxima over the layers gives the value for every at-most budget. Predecessors recover the attaining books. The graph and layer tables each have $O(N^2)$ entries. Each entry is a sum of at most N rational edge weights of polynomial input encoding length, proving the bit claim. Theorem 5.3 and the canonical response reconstruct original lotteries and service at $t_j = b_j$. $\square$

The implementation forms this frontier in one count-indexed pass. It then exports each budget's state table and original policy in the existing deficit-certificate format. The independent checker reconstructs full-capacity kernels and checks all Bellman inequalities without importing the frontier solver. For positive deficit, the approximate state count is $\lfloor 2Km(\bar{b} - B)/\varepsilon \rfloor + 1$; it depends on the unused obligation capacity, not on the number of distinct caps. The exact saturated frontier and the positive-deficit guarantee thus describe adjacent but different computational regimes.

8.5 Exact lattice optimization at arbitrary command budgets

Zero service cost is a substantive subclass of the original model, not a relaxation of its obligation or realization constraints. It includes the weak SUBSET SUM construction. The deficit representation yields an exact pseudo-polynomial algorithm for that subclass even with heterogeneous terminal rewards.

Theorem 8.6 (Exact zero-service lattice algorithm). *Suppose $h_j \equiv 0$ and the catalog, probabilities, caps, promise, charges, and terminal reward values at catalog commands are rational. Rewards are increasing and concave, with a common catalog and ceiling-induced prefix eligibility. Let D be a positive integer such that*

$$DB \in \mathbb{Z}, \quad D\pi_j b_j \in \mathbb{Z}, \quad D\pi_j a_i \in \mathbb{Z} \quad \text{for every } j, i. \quad (34)$$

Set $R = \bar{b} - B$. The merged-origin recurrence with mesh $1/D$, terminal index exactly DR , and no approximation allowance solves the original problem exactly for every $1 \leq m \leq N$. For rational quadratic rewards its arithmetic bound is (33) with $Q = DR + 1$. The algorithm is pseudo-polynomial in numerical DR and polynomial in the other parameters and their encoding lengths. It is not asserted to be FPT in m alone or polynomial in $\log D$.

Proof. Condition (34) implies that every weighted terminal component capacity $\pi_j[\min(b_j, v) - u]_+$ and every service capacity $\pi_j(b_j - u)_+$ belongs to $D^{-1}\mathbb{Z}$. It also implies that every arc capacity, every potential, every anchor, and R belong to that lattice. In the zero-cost case each Φ_e is concave piecewise linear: its marginal segments are the sorted positive terminal chord slopes followed by a zero-slope service segment. Every segment length is a sum of weighted component capacities, hence a lattice multiple. The deficit kernel, obtained by reversing the resource coordinate, has the same breakpoint property. Centering adds a common linear term without changing these breakpoints.

Fix a feasible path. Maximizing its separable concave piecewise-linear deficit payoffs subject to one exact sum can be done by filling marginal segments in decreasing order. Ties may be ordered to respect the segment order within each edge. Every full segment length and the total R are integer multiples of $1/D$. Therefore the possibly partial final segment also receives a lattice amount. Some optimum on that path has every deficit on the common lattice. Maximizing over the finite path set proves that the exact lattice recurrence contains a global optimum. Its selected book is allocated at the original promise by the canonical history-level construction; the resulting lower and upper values are equal as rational numbers.

The arithmetic count follows from Proposition 8.4; no independent anchor loop is introduced. A sufficient D is the least common multiple of the denominators in (34). Its bit length is at most the sum of those denominator bit lengths, but its numerical value can be exponential in that sum. Hence enumerating

$DR + 1$ grid positions is pseudo-polynomial, not a binary-input polynomial algorithm. This distinction explains compatibility with the weak hardness result. $\square$

For equal probabilities $1/k$, commands and caps on a grid of step $1/H$, and a promise on a grid of step $1/(kH)$, one can take $D = kH$. This gives an explicit operationally interpretable scaling parameter: history frequency and commanded-quantity granularity. Arbitrary rational probabilities can instead generate a much larger least common multiple. Nonzero quadratic service costs introduce interior stationary allocations that need not lie on this input lattice, so their exactness is not inferred from Theorem 8.6; the centered additive theorem and exact price search continue to cover them.

Hexadecimal proof serialization changes neither the lattice nor this complexity classification. A fraction with a twenty-thousand-bit denominator still requires those bits to be stored and processed. The implementation records numerator and denominator bit lengths, uncompressed and compressed certificate bytes, and explicit state-limit failures. It does not disable Python’s decimal-integer conversion limit, and it does not interpret successful serialization as numerical tractability.

For fully arbitrary rational inputs, “pseudo-polynomial” here refers to the scaled integral encoding in (34), not to the largest individual denominator before scaling. A least common multiple of many small denominators can itself be large. Under the common-grid normalization just specified, the conventional numerical parameter is H and $D = kH$ suffices. That subclass contains the SUBSET SUM construction and has a genuine pseudo-polynomial algorithm in its explicit granularity. No strong-hardness classification of the unrestricted quadratic-service model is inferred from this result.

9 Antecedent Computational Evidence

The following study is retained from the complete predecessor revision. Its protocols, source hashes, raw records and unsuccessful runs are unchanged; its timings are not pooled with the new executions in Section 10. Its original article tables are reproduced in the electronic companion.

9.1 Protocol and limits

The fixed study contains 30 cases and 142 applicable method runs. One-factor sweeps vary histories through 768, catalog size through 65, command budget through 12, absolute accuracy through 10^{-6} , and opening-charge scale. Four cases have heterogeneous terminal rewards; six are the original difficult historical models. Caps, probabilities, service curvatures, and ceilings are heterogeneous. Repeated base cases in different sweeps are intentional paired comparisons, not independent statistical observations. Table EC.5 reports every new price-path case; Table EC.6 gives the paired historical reruns.

Each method executes in a fresh process with one thread, a four-second algorithm-time alarm and a 2-GiB address-space cap. Imports are timed separately; peak resident memory is measured rather than assumed equal. The mixed-integer reference uses the original support-indicator formulation with 32-segment tangent and secant service envelopes and a two-second solver limit for each call. Total construction and both solves remain subject to the common alarm. A native solver can delay delivery of a Python signal; a separate

five-second algorithm-phase process limit records that outcome. This distinction, actual elapsed time, and every limit event are retained.

The baselines are the inherited uniform-resource grid, exact book enumeration, class-mean box subdivision, and the direct numerical mixed-integer model. The inherited grid and box implementations support common rewards only and are not run on the four heterogeneous-reward cases. The price method makes at most 12 price attempts per node and evaluates at most 255 nodes. Rational proof checking is separate, with a 60-second verification alarm. An interrupted old routine is not assigned its unfinished computations as a proof: only an independently feasible incumbent and a universal upper bound survive unless the routine has returned a checkable certificate. Numerical mixed-integer status is never interpreted as rational certification. The full limits and input hashes were committed before these runs.

9.2 Scaling, accuracy, and unsuccessful cases

Across the original study, price paths produce 10 exact closures, 19 positive-width tolerance completions, and 1 resource-limited outcomes. All 30 resulting price certificates pass independent verification. Table EC.7 retains every method’s outcome. The larger-catalog cases demonstrate performance beyond what the allotted exhaustive enumeration can complete; they do not establish polynomial worst-case exact search. The largest history case illustrates the remaining limit. Its enclosing interval is reported even when it misses the requested tolerance.

The historical comparison is no longer between newly timed code and a different publication environment. All five applicable methods are rerun under the same caps. Enumeration remains attractive on these small catalogs, and numerical mixed-integer solutions can be fast. The benefit of the new path oracle is a substantially smaller exact-rational proof workload than the uniform resource grid on these cases, not a universal ordering of solver times. Compressed price certificates in the full study range from 541 to 5895 bytes. Table EC.5 also reports checking time and peak process memory; small proof files do not imply zero verification cost.

Ten separate zero-width runs expose the search’s anytime behavior in Table EC.8. They include all six historical models and the largest two catalog and budget configurations. Of these, 10 close exactly within the stated limits. Every per-oracle and per-node checkpoint remains in the execution record. Unlike a uniform grid, a smaller requested tolerance does not necessarily require more resource states: a supporting price or a completed discrete partition can close an interval exactly. The construction’s worst-case guarantee remains finite exponential search, as stated in Theorem 6.1.

After deriving the complexity theorem, we separately froze four SUBSET SUM challenge configurations before executing them. Their outcomes occupy the lower panel of Table EC.7. These are deliberately difficult structural cases, not representative operational inputs. The source integer weights are even and the target odd; integer subset-sum dynamic programming supplies their exact reference distance. The new method also encounters node and time limits on this family. These failures are informative: the price bound does not remove the discrete obstruction proved in Theorem 4.2. Grid and box comparisons use width $1/(8K)$, while price and enumeration request exact closure, as explicitly declared in the addendum; their success counts are not pooled as identical-accuracy outcomes.

9.3 Screening and independent correctness checks

Table EC.9 reports all 24 screening cases, crossing eight nonuniform-cap models and charge multipliers 0.9, 1, 1.1 around the target-box envelope. The catalogs include several possible low anchors and a mixture of retained and deleted commands; promises and budgets vary. Exact inclusion and exclusion optima identify useful, tied, and strictly irrelevant candidates. Forced price bounds remove 45 possible-anchor occurrences across these cases. Rates, missed strictly irrelevant commands, wholly ineligible removals, and every charge slack are recorded. The stronger test incurs incumbent-search and bound-checking costs; no generic preprocessing speedup or removal guarantee is inferred. Three additional sharpness configurations make the same command necessary below the exclusion threshold, tied at equality, and excluded above it.

The regression suite verifies 130 models, including 60 common-cap models and 30 heterogeneous-reward models. All 130 global price intervals close at the exhaustive optimum. Independent singleton/pair certificates pass on the 60 common-cap cases, 40 of which have more than one eligibility class. The suite checks 2,644 fixed books, 390 restricted-price oracles, 2,035 nested-layer reconstructions, and the heterogeneous component construction and optimal component equality on all 2,644 books. A deliberate positive root-duality-gap example requires branching; a one-node interruption retains a valid incomplete-width certificate. Nine altered certificates are rejected. The separately declared hardness validation checks all 1692 optional subsets on 60 reductions and all 3004 feasible catalog books in the smaller reductions. Every distance identity and yes/no equivalence agrees. These finite checks verify implementation consistency; the universal statements rest on the proofs.

10 Current Computational Evidence

The current study evaluates the new heterogeneous resource implementation and exact lattice regime, and measures the search and proof costs of price paths and forced-command screening. It does not pool the preceding section’s R54 timings with new executions. The primary protocol specifies 186 timed method runs. An extension inherited from post-inspection development adds 39 exact-target and encoding runs; nine untimed mechanical replays supply search diagnostics. The present revision freshly executes all 225 runs from a source snapshot committed before execution. Its cases, methods, limits, and tolerances match the recovered corrected design. These are new execution records, not a relabeling of unpublished local timings. All failures are retained, and every numerical statement below is generated from the new records. The companion documents the earlier publication failure and the distinction between recovered development and current evidence.

10.1 Design and outcome definitions

The primary comparison crosses three seeds, two optimization limits (0.5 and 2 seconds), and four model families. Their history, catalog, and budget dimensions are (6, 7, 3), (8, 9, 3), (24, 17, 4), and (96, 25, 4). The first family has common rewards and the others have heterogeneous rewards. Probabilities are uniform; caps, ceiling increments, service curvatures and reward parameters are drawn on declared rational grids. The

promise is three quarters of aggregate capacity. Charges are nonnegative, and all numerical parameters and instance digests are saved before the corresponding method executes. The same model at different limits is a paired observation, not a new independent instance.

Each method runs in a fresh process with a two-GiB address-space limit and single-thread numerical libraries. The price and deficit methods request width $1/20$ in this panel. Enumeration requests exact completion. The direct support-indicator mixed-integer comparator uses numerical cost envelopes and has a separate evidence status. Setup, optimization, serialization and independent checking are timed separately; a twelve-second initial checker allowance is not part of the optimization allowance. Native-library overruns, failures and the last completed phase remain reported. Table 1 gives all family-level medians and success counts.

10.2 All declared runs and certificate costs

The source-frozen execution retains all 225 timed records: 88 exact; 23 numerical; 2 state-limit; 29 time-limit; 83 tolerance. Status counts summarize record completeness, not a pooled success rate across unequal accuracies, budgets, or method definitions. The separate panels in Tables 1–4 retain their original targets. 200 runs produce rational certificates, including conservative fallback intervals from interrupted algorithms. 200 certificates have a successful independent check. 3 initial checks reach their twelve-second allowance; 3 unchanged certificates subsequently pass the separately timed extended check. No initial status is rewritten. The longest extended check takes 36.358 seconds. Numerical MIP returns remain separate from exact rational evidence.

The heterogeneous deficit method completes 27 of 30 requested primary and accuracy runs. 27 completed runs have an independently checked exact reference on the identical model. The maximum measured incumbent loss among those comparisons is 0.00111898. This observed loss is not the guaranteed upper-minus-lower interval. Interrupted resource computations and expensive independent checking remain part of the record; the faster convolution theorem does not make every complete proof pipeline inexpensive.

10.3 Exact-target search and end-to-end screening

The exact-target extension contains 18 price runs, of which 17 close at zero rational width. Table 2 reports all nine two-second cases, with nodes, oracle calls, root gaps, fixed-book comparisons, depth and elapsed optimization time. The half-second counterparts remain in the raw records. Nine separate deterministic replays expose each tested price and required/forbidden set. They measure search mechanics, not replacement benchmark times. Across recorded price runs, maximum tree depth is 3 and the largest completed-oracle count is 45. Retained prices use at most 21 numerator bits and 20 denominator bits. The original R54 hard reduction instances, including failures, remain unchanged in the historical evidence. Neither root closure nor these observed bit lengths imply a polynomial worst-case search tree.

There are 18 paired screening comparisons under the same two-second total optimization allowance. 0 pairs improve total optimization-plus-serialization-plus-checking time; the median paired total-time ratio is 1.604. Table 3 reports charges, deletions by eligibility category, final widths and both total times. Screening

is charged for incumbent construction and forced-command bounds before the reduced solve. All unfavorable pairs remain visible. A smaller catalog or tighter bound is not itself an end-to-end speed improvement.

10.4 Lattice scaling and implementation checks

Among the eighteen zero-service lattice cases, 18 return an independently checked exact optimum. Cap-grid denominators are 8, 16 and 32; command budgets are two and four; all three seeds are retained. The table reports the actual resource-grid size, which also depends on probabilities and commanded-quantity granularity. The separate encoding stress uses denominators 2^8 , 2^{16} and 2^{32} and preserves any state-limit rejection rather than assigning it an optimum.

The exact regression covers 36 general models, 36 lattice models, 36 common-reward cap-count constructions, four strict $2q$ examples, and 200 convolutions with support holes. The joint cap-reward extension adds 36 genuinely heterogeneous reward models, each with two joint classes and independently checked original-policy reconstruction. Seven certificate mutation classes are rejected. A full twenty-thousand-bit rational certificate and a pre-allocation enormous-state rejection distinguish correct serialization from numerical tractability. These finite tests validate implementations; the universal assertions rely on the stated proofs. All operational data remain synthetic.

11 Conclusion

Paid finite-catalog renewal design admits exact selected-boundary resource and price representations under its original expected and realized constraints. History-level component packing extends the primal identity to heterogeneous rewards. The representations separate model-specific structure from established resource allocation, Lagrangian bounding and discrete search mechanisms.

The cap-count theorem identifies a tight small-menu structure: twice the number of distinct expected caps is both sufficient and sometimes necessary under a common reward. The deficit coordinate makes the terminal resource state independent of the anchor, removes a catalog factor from the primal approximation, and supports an exact zero-service lattice algorithm at arbitrary command budgets. These results complement, rather than replace, the common-cap and common-prefix exact subclasses, the weak encoding-sensitive hardness construction, safe catalog rules and robust capacity-repair arguments.

The computational records distinguish the costs of optimization, proof representation and independent checking. A positive-tolerance price bound may stop at the root; exact-target trajectories reveal when branching is actually needed. The heterogeneous deficit implementation supplies a polynomial alternative, but its explicit checker can cost substantially more than optimization. Screening can improve a bound or reduce the book universe without improving total proof time. These distinctions identify which structural result is useful for a given instance instead of treating one completion label as a general performance claim.

The service-credit architecture supplies an implementation and calibration interface, not a claim of field validation. Fixed feasibility controls and estimated payoff primitives play different roles. Keeping them separate permits original-instance certificates and explicit payoff-error transfers to inform operational decisions without altering the accepted obligation.

Code and Data Accessibility

The accompanying code and data retain the complete predecessor source and evidence package and provide the current heterogeneous deficit optimizer, lattice mode, exact price search, book enumeration, numerical mixed-integer comparison, independent checkers, input records, full trajectories, and every declared failure. Current and antecedent execution records have separate protocols and source hashes. A post-inspection exact-target study and a separate extended-verification pass are explicitly identified. Compressed certificates include the requested model; external instance hashes bind verification to that model. Reproduction instructions, the reader map, provenance, and build validation are supplied with the package. Repository transport and formatting checks are not scientific contributions or editorial acceptance decisions.

Approximate reward types provide a certified tradeoff between menu size and value without perturbing feasibility. Saturation provides a separate exact all-budget regime with positive service costs. Together these results show why the useful structural parameters are not interchangeable with numerical precision or command count alone.

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Table 1: Matched positive-tolerance runs: median optimization seconds and successes out of three seeds.

Family	k, N, m	Limit	Price	Deficit	Enumeration	MIP
Common	6, 7, 3	0.5	0.004 [3]	0.020 [3]	0.013 [3]	0.010 [3]
Common	6, 7, 3	2	0.006 [3]	0.021 [3]	0.013 [3]	0.009 [3]
Heterogeneous S	8, 9, 3	0.5	0.008 [3]	0.040 [3]	0.028 [3]	0.028 [3]
Heterogeneous S	8, 9, 3	2	0.008 [3]	0.041 [3]	0.029 [3]	0.028 [3]
Heterogeneous M	24, 17, 4	0.5	0.040 [3]	0.260 [3]	0.499 [0]	0.040 [3]
Heterogeneous M	24, 17, 4	2	0.039 [3]	0.261 [3]	2.004 [0]	0.040 [3]
Heterogeneous L	96, 25, 4	0.5	0.304 [3]	0.516 [0]	0.505 [0]	0.435 [2]
Heterogeneous L	96, 25, 4	2	0.309 [3]	0.929 [3]	2.006 [0]	0.436 [3]

All price and deficit targets are $1/20$; enumeration requests exact completion. MIP entries count returned numerical results, not rational certificates. Limits are optimization seconds per fresh process. Setup, serialization and independent checking are measured separately. Verification timeouts and subsequent checks of unchanged certificate bytes are reported separately in the generated execution summary.

Table 2: Exact-target extension: all nine models at the two-second optimization limit.

k, N, m	Seed	Root gap	Nodes/calls	Depth	Books	Enum. L	Price V^*	Seconds
8, 9, 3	5601	0	1/4	0	4	0.4289	0.6580	0.012
8, 9, 3	5602	0.00181	7/45	3	6	0.8899	0.9991	0.087
8, 9, 3	5603	0.00335	5/43	2	10	0.6314	0.6973	0.073
24, 17, 4	5601	0	1/3	0	3	0.2490	0.9290	0.069
24, 17, 4	5602	0	1/4	0	4	0.2717	0.9231	0.081
24, 17, 4	5603	0	1/2	0	2	0.1073	0.8537	0.055
96, 25, 4	5601	0	1/3	0	3	0.1317	0.9461	0.514
96, 25, 4	5602	0	1/2	0	2	0.0268	1.0258	0.423
96, 25, 4	5603	0	1/2	0	2	0.0281	0.9426	0.399

Root gap is the remaining rational gap after the root price attempts, before any split. Nodes are evaluated search nodes; calls are completed priced-path oracles. Books count distinct original fixed books in the untimed mechanical replay. Enumeration L is the feasible value of lexicographic enumeration stopped at that same book count, not an equal-work or equal-time comparison. The raw record states each final interval, status and independent-check result; the generated text reports the actual number of exact closures. The half-second counterparts and complete traces remain in the raw records.

Table 3: Screening economics: medians over three seeds at the same two-second total optimization allowance.

k, N	Charge	Removed	A/I/O	Direct total	Screen total	Ratio	Width change
8, 9	0	6	4/0/13	0.018	0.027	1.52	0.00244→0
8, 9	1/100	5	3/0/14	0.021	0.035	1.53	0.00293→0.00546
8, 9	1/10	6	3/0/16	0.019	0.029	1.53	0.00604→0.00604
24, 17	0	12	4/0/27	0.109	0.196	1.81	0.000173→0.000122
24, 17	1/100	14	6/0/35	0.114	0.188	1.67	0.00126→0
24, 17	1/10	14	6/0/37	0.125	0.206	1.64	0→0

Totals include optimization, serialization and independent checking, but not process setup. The screening allowance includes seed construction, forced bounds and reduced-catalog search; it is not added to a fresh full solve allowance. A/I/O are total removals across the three seeds: possible anchors, wholly ineligible commands, and ordinary nonanchors. Ratios are medians of paired total-time ratios. The number of actual paired time improvements is reported in the generated execution summary; a stronger bound is not itself a time saving.

Table 4: Exact lattice results and explicit encoding limits.

Cap-grid H	Budget	Seeds	Exact	Median Q	DP seconds	Enum. seconds
8	2	3	3/3	22	0.0121	0.0044
8	4	3	3/3	22	0.0188	0.0247
16	2	3	3/3	40	0.0196	0.0042
16	4	3	3/3	40	0.0335	0.0219
32	2	3	3/3	73	0.0328	0.0027
32	4	3	3/3	73	0.0623	0.0171
Post-inspection encoding stress: six histories, seven commands, budget two.						
2^8	2	1	EXACT		0.1824	—
2^{16}	2	1	STATE LIMIT		0.0010	—
2^{32}	2	1	STATE LIMIT		0.0011	—

All lattice runs request exactly zero width; their comparator is exact book enumeration. H describes the cap grid, while the implementation computes a common denominator for both cap and command components. The two larger stress grids exceeded the declared 600,000 Bellman-state allowance and return no optimality interval. Successful hexadecimal serialization does not remove this numerical-size barrier.